Important Topics from all modules

UNIT 1: OOP Concepts and Overview of Java

Important Topics for 5 Marks

OOP concepts: Class, Object, Abstraction, Encapsulation
Difference between OOP and Procedural Programming
Java data types and variable
Operators in Java
Java methods and method overloading
Arrays in Java
Scanner vs BufferedReader
Control structures (selection and looping)

Important 10-Mark Topics

OOP principles with suitable examples
Comparison between Procedural and OOP approaches
Java control structures and methods with examples
Arrays and method overloading with program illustrations
Input/output operations using Scanner and BufferedReader

UNIT 2: Objects and Classes

Important 5-Mark Topics

Constructors and their types
Finalizer in Java
Visibility modifiers
this reference
String and Character classes
StringBuffer and StringBuilder
Methods and objects

Important 10-Mark Topics

Constructors and object creation process
Visibility modifiers with accessibility comparison
String, StringBuffer, and StringBuilder comparison
Methods, objects, and this reference
Built-in classes and their applications

UNIT 3: Inheritance, Polymorphism and Package

Important 5-Mark Topics

Types of inheritance in Java
Super class and sub class
Method overriding
Dynamic binding
Object class
instance of operator
Abstract class
Interface
Package creation and importing packages
Access specifiers in packages

Important 10-Mark Topics

Inheritance and method overriding with examples

Abstract class vs Interface

Polymorphism and dynamic binding

Package creation, importing, and member access

Complete discussion on inheritance, interfaces, and packages

UNIT 4: Exception Handling and Multithreading

Important 5-Mark Topics

Exception handling basics

try-catch-finally blocks

throw and throws keyword

Checked and unchecked exceptions

User-defined exceptions

Thread life cycle

Thread priorities

Thread synchronization

Single-tasking vs Multi-tasking

Important 10-Mark Topics

Complete exception handling mechanism in Java

Types of exceptions and user-defined exception

Multithreading and thread life cycle

Thread synchronization and priorities

Creation, execution, and termination of threads

UNIT 5: Event Handling, GUI and Collections

Important 5-Mark Topics

Event handling in Java

Event types

Mouse events and key events

Panels and Frames

FlowLayout

BorderLayout

GridLayout

Swing basics

ArrayList, LinkedList, and Vector

GUI components (Button, Label, TextField, Checkbox, RadioButton)

Important 10-Mark Topics

Event delegation model

Layout managers comparison (FlowLayout, BorderLayout, GridLayout)

GUI components and their applications

AWT vs Swing

Collection framework overview (ArrayList, LinkedList, Vector)

Designing a simple GUI application